Independent Quantitative Audit and Strategic Roadmap: The Aegis-Finance Project

The following report constitutes an exhaustive, adversarial, and independent quantitative evaluation of the Aegis-Finance project. The analysis treats all internal documentation, assertions, and backtested metrics as falsifiable claims. By systematically attacking the validation arithmetic, the process controls, the data infrastructure, and the core theoretical thesis, the objective is to expose statistical artifacts, operational look-ahead biases, and capital-allocation flaws before they contaminate live execution. The evaluation leverages advanced econometric frameworks, recent empirical literature in asset pricing, and state-of-the-art open-source quantitative tooling to deliver a definitive roadmap for the project's next 90 days.

1. Validation of Empirical Results and Statistical Arithmetic

The central claim of the project is the identification of a limited number of "surviving" signals extracted from a pool of 179 candidates, alongside a live paper-trading record of 55 days[1]. A rigorous audit of the underlying mathematics and logic reveals critical vulnerabilities in how statistical significance, multiple testing, and data integrity have been handled.

1.1 The Multiple-Testing Arithmetic and the False Strategy Theorem

VERIFIED: The project utilizes a Deflated Sharpe Ratio (DSR) and a multiple testing hurdle of $t \geq 3.0$ derived from Harvey, Liu, and Zhu (2016)[2]. However, the application of this arithmetic contains a fundamental mismatch: the production ship-gate deflates against 18 registered trials, while the true search space encompasses 179 evaluated candidates[1].

Applying the False Strategy Theorem (Bailey & López de Prado, 2014) using Extreme Value Theory demonstrates the severity of this discrepancy. Under the null hypothesis of zero skill, the expected maximum Sharpe Ratio ($\widehat{SR}_{max}$) from a pool of N independent strategies is approximated by:

$$E\left[\max_n \widehat{SR}_n\right] \approx E[\{\widehat{SR}_n\}] + \sqrt{V[\{\widehat{SR}_n\}]}\left[(1-\gamma)\Phi^{-1}\left(1-\frac{1}{N}\right) + \gamma\Phi^{-1}\left(1-\frac{1}{Ne}\right)\right]$$

Where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant. For a 15-year explore window, the standard error of an annualized Sharpe ratio is approximately $1/\sqrt{15} \approx 0.258$.

Evaluation Cohort	N (Candidates)	Expected Max Null Sharpe	95th Percentile Null Sharpe	Required t-statistic Hurdle
Machine Registry	18	0.48	0.65	≈ 1.86
Documented Ledger	179	0.71	0.89	≈ 2.73
Latent Search Space	500+	0.79	0.98	≈ 3.05

VERIFIED: Deflating against $N = 18$ artificially lowers the survival bar. At $N = 179$, a strategy exhibiting a backtested Sharpe of 0.70 is mathematically indistinguishable from pure noise[1]. The survivors—gp-small, fusion, and TSMOM-XA—must be immediately re-evaluated against the true candidate count.
REPORTED: Furthermore, if parameter variations, lookbacks, and universe permutations within the 179 candidates are accounted for, the effective N likely approaches 500–1,000, aligning exactly with the recent reaffirmation by Harvey, Sancetta, and Zhao (2026) that the minimum valid t-statistic cutoff in factor models remains at least 3.0 to control the False Discovery Rate (FDR)[5]. The literature emphasizes that ignoring unpublished or latent tests invalidates the control of FDR, confirming that the ledger of 179 is the absolute floor for the multiple testing adjustment[6].

1.2 The Placebo-Gate Logic and Pseudoreplication

VERIFIED: The project utilizes a random-date placebo gate to control for selection bias and cohort drag, notably in the evaluation of 13D/13G filings[1]. The architecture is conceptually sound, but the statistical implementation is fatally misspecified due to pseudoreplication.
The documentation indicates that the pooled placebo test aggregates five random seeds into a single dataframe, calculating a clustered t-statistic across the inflated sample size. This approach multiplies the number of observations by five while analyzing the exact same underlying cohort of firms. Because the t-statistic scales mechanically with $\sqrt{N}$, treating five stacked replicates as independent scales the test statistic by approximately $\sqrt{5} \approx 2.23$ [1]. Consequently, a marginal seed-level effect is artificially amplified into a "statistically

significant" placebo failure.

SPECULATION: The seed-level t-statistic of -4.06 (versus the pooled -3.17) is actually stronger evidence of a cohort drag, but the pooling mechanism itself remains arbitrary and unregistered[1].

To correctly distinguish between cohort drag and true signal without inflating degrees of freedom, the placebo generation must employ a strict permutation test or a circular block bootstrap across time. Shuffling the event dates *across* permnos while preserving the exact calendar-month marginal distribution ensures that macroeconomic timing and cross-sectional dependence are not destroyed during the randomization process.

1.3 Data Leaks and Forward Performance Illusions

VERIFIED: The internal audit exposed critical data leaks, including the use of FRED reference-date macroeconomic data rather than publication-date vintages, injecting up to five months of look-ahead bias into the crash prediction models[1]. This completely invalidates the reported walk-forward Area Under the Curve (AUC) and Brier scores. Furthermore, the quantile_return_spread function generated a fabricated +174 bps spread out of a constant factor by breaking ties alphabetically via row order[1].

REPORTED: The 37-day (55 calendar days) live paper-trading record across ten lanes is an epistemic illusion. The standard error of an annualized Sharpe ratio over 55 trading days (0.218 years) is roughly $1/\sqrt{0.218} \approx 2.14$ [1]. It takes a minimum of 16 years of out-of-sample trading to prove an Information Ratio (IR) of 0.5 at a $t = 2$ significance level[1].

The current forward window provides utility solely as an implementation audit—verifying execution logic, cost models, and data pipeline stability. It cannot be used to make any performance claims.

2. Evaluation of Process Controls and Epistemic Rigor

The gap between stated discipline and machine-enforced discipline threatens the project's integrity. While the pre-registration protocol and negative-results ledger represent elite practices, operational loopholes undermine the process.

2.1 The Explore/Confirm Wall and Multiple Comparisons

REPORTED: The split between the explore (2004–2018) and confirm (2019–2024) periods is compromised if the confirm dataset is accessed more than once per strategy family[1].

Repeatedly tuning a model on the explore set and validating it on the confirm set rapidly transforms the holdout data into an extension of the training set.

To systematically penalize this, the framework must integrate the **Romano-Wolf Stepdown Procedure**[7]. This algorithm controls the Family-Wise Error Rate (FWER) by utilizing multiplier bootstrap techniques to account for the dependence structure across all tested hypotheses. Implementing the rwolf equivalent in Python (e.g., via the DoubleML package) ensures that the statistical significance of any "surviving" candidate is adjusted for every previous query against the confirm set[8].

Additionally, comparing multiple models out-of-sample requires **Hansen's Superior Predictive Ability (SPA) test** (2005)[12]. A stationary block-bootstrap approach must be used to preserve the autocorrelation and volatility clustering inherent in financial time series[12]. Available Python repositories (e.g., RSv618/superior-predictive-ability) allow for direct computation of SPA, calculating whether any candidate strategy genuinely outperforms a benchmark after correcting for data-snooping[16].

2.2 Over-Disciplining and the Rademacher Anti-Serum

SPECULATION: The project currently penalizes volatility and drawdowns using traditional optimization methods that often succumb to parameter instability. By demanding strict performance across arbitrary sub-samples, the pipeline may be over-disciplining, killing true signals with controls that possess more power against truth than against noise.

REPORTED: To prevent over-fitting during the tuning of risk-management parameters, the architecture should incorporate **Rademacher Complexity**[18]. The **Rademacher Anti-Serum (RAS)** methodology, proposed by Paleologo (2025), adjusts empirical Sharpe ratios by introducing Rademacher random vectors to measure how easily a strategy class aligns with pure noise[18]. By calculating the Rademacher Complexity of the strategy space, the algorithm generates a RAS-adjusted Sharpe Ratio[18]. Implementing this via the RSv618/rademacher-anti-serum repository acts as an objective, mathematical penalty for strategy flexibility, preventing over-disciplining while strictly guarding against curve-fitting[17].

2.3 Required Protocol Adjustments Ranked by Epistemic Payoff

1. **Machine-Enforced Trial Counters:** The true trial count ($N = 179$) must be a machine-readable artifact that permanently binds the DSR and PBO (Probability of Backtest Overfitting) gates, correcting the CANON violation where the registry currently shows 18 adoptions and 0 rejections[1].
2. **Implementation of Romano-Wolf and Hansen's SPA:** Replace naive out-of-sample t-tests with FWER-controlled stepdown procedures and block-bootstrapped SPA evaluations to formally account for test dependence[21].
3. **Rademacher Complexity Penalties:** Integrate RAS to dynamically penalize highly parameterized strategies, allowing for the salvage of models that exhibit consistent predictive power without succumbing to parameter tuning[18].

3. Authoritative Sources, Repositories, and Point-in-Time Infrastructure

The project is heavily constrained by the decision to build bespoke infrastructure when mature, peer-reviewed, open-source alternatives exist, and by failing to fully exploit licensed, high-fidelity datasets.

3.1 Point-in-Time Data via WRDS and CCM

VERIFIED: The utilization of the WRDS subscription must move from optional to mandatory. Relying on Yahoo Finance limits the universe to surviving entities, invalidating cross-sectional anomalies[1].
REPORTED: The CRSP/Compustat Merged Database (CCM) provides the foundational crosswalk between Compustat fundamental data (GVKEY) and CRSP market data (PERMNO)[24]. Attempting to merge via CUSIP introduces severe matching degradation due to historical CUSIP recycling and OTC market exclusions[24].
To safely map fundamental data to market data, the system must utilize the crsp.ccmxpf_lnkhist table. By enforcing the condition linkdt <= date <= linkenddt, the database naturally constructs a survivorship-free, point-in-time universe[24]. The linking process must also handle consecutive date ranges where LINKPRIM flags change between 'P' (Primary) and 'C' (CRSP-assigned), ensuring continuous entity tracking[25].
Furthermore, the currently downloaded 109 MB of BoardEx data is truncated exactly at 500,000 and 1,000,000 rows, indicating a SQL LIMIT truncation without an ORDER BY clause, resulting in a non-random, heavily clustered subgraph[1]. This dataset must be completely re-pulled.

3.2 High-Value Git Repositories

Rather than hand-coding portfolio optimizers and signal validations, the following repositories should be integrated:

Repository	**Core Utility**	**Specific Application to Aegis**
OpenSourceAP/CrossSection	319 replicated firm-level characteristics[29].	Every new signal must be orthogonalized against this pre-existing universe to prove marginal novelty[30]. Note the critical AnnouncementReturn look-ahead bug patched in v1.4.1[29].
skfolio	Portfolio optimization built on scikit-learn[34].	Replaces naive optimizers. Supports Hierarchical Risk Parity (HRP), distributionally robust optimization, and built-in cross-validation (GridSearchCV) for

		hyperparameter tuning[34].
vectorbt	Vectorized backtesting via Numba and NumPy[37].	Accelerates the hot path, simulating thousands of parameter variations simultaneously[37]. Combines well with Rademacher Anti-Serum to offset backtest overfitting[18].
microsoft/RD-Agent	Multi-agent framework for data-centric factor/model co-optimization[40].	Automates the Research and Development stages using LLMs and a code-generation agent (Co-STEER). Demonstrates up to 2x higher annualized returns using 70% fewer factors[40].

REPORTED: The microsoft/RD-Agent framework represents the direct competitor to the Aegis autonomous lab. It utilizes a multi-armed bandit scheduler to test factor hypotheses, evaluating experimental outcomes and informing subsequent iterations through a closed-loop cycle[42]. Integrating the logic of the Synthesis and Implementation units from RD-Agent could vastly accelerate the pipeline's signal generation capabilities[42].

4. Idea Generation, The "Social Bias" Thesis, and Capital Allocation

The core thesis surrounding social biases, narrative context, and network effects is conceptually profound and well-supported by modern financial economics. However, its translation into executable signals requires a strict delineation between compensated risk and uncompensated noise.

4.1 Steelmanning Social Bias and Networks

REPORTED: The premise that "profitable-but-unfashionable companies are systematically undervalued" is corroborated by robust academic literature:

- **The Neglect Premium:** Fang and Peress (2009) establish that stocks with zero media coverage earn higher future returns than heavily covered equivalents, controlling for size and momentum[1].
- **Analyst Coverage Decline:** Scherbina (2008) documents that a decline in analyst coverage (ChNAnalyst) acts as a powerful predictor of neglect, yielding cross-sectional spreads[46].
- **Political and Social Networks:** Cooper, Gulen, and Ovtchinnikov (2010) show that

political contributions predict returns[1]. Wei and Zhou (2025) specify that congressional trading outperformance is heavily conditioned on *leadership* status, rather than general congressional trades[1].
- **Network Centrality:** Cohen, Frazzini, and Malloy (2008, 2010) prove that educational and board interlocks significantly predict returns[1].

SPECULATION: Combining high gross profitability (GP) with deteriorating analyst coverage (ChNAnalyst) represents a highly potent, testable hypothesis that perfectly models the "profitable but unfashionable" thesis.

4.2 Attacking the Weak Links: Capacity and Short-Leg Dominance

VERIFIED: The evaluation reveals that a vast majority of cross-sectional spreads (e.g., 99.9% of institutional ownership anomalies and 88% of skewness anomalies) reside entirely within the short leg, which the long-only retail book structurally cannot capture[1].

REPORTED: This outcome aligns with Drechsler and Drechsler (2014), who demonstrate that the major cross-sectional anomalies effectively disappear within the 80% of stocks that have low short fees, and are greatly amplified among those with high fees[1]. Similarly, Hou, Xue, and Zhang (2020) replicate 452 anomalies and find that with microcaps mitigated via NYSE breakpoints and value-weighted returns, 65% fail to clear a $t = 1.96$ hurdle, and 82% fail at $t = 2.78$ [47].

Therefore, the pipeline must apply an **ex-ante borrow fee and short-interest eligibility filter**. If an anomaly fundamentally relies on high-borrow-fee or heavily shorted microcaps to generate its spread, it must be rejected as uninvestable at retail scale[1].

4.3 Extending the Thesis: LLM Extraction and the Distraction Effect

REPORTED: The deployment of Large Language Models (LLMs) in quantitative finance faces severe look-ahead and memorization hurdles. Glasserman and Lin (2023) identify a critical "distraction effect," wherein an LLM's generalized knowledge of a high-profile company overrides the specific sentiment of the text being analyzed, causing performance decay out-of-sample[50]. The authors demonstrate that anonymizing entities (removing the company name) prior to LLM analysis significantly improves signal reliability, as the distraction effect frequently has a greater negative impact than standard look-ahead bias[51].

Furthermore, NBER w33344 (Ludwig, Mullainathan, and Rambachan, 2025) outlines the exact boundary condition: LLMs must be used for *estimation* (extracting predefined data from text) rather than *prediction*[54]. Applying this logic requires:

1. Using deterministic tools (cosine similarity, TF-IDF) for baseline textual novelty.
2. Deploying LLMs with strict, temperature=0, enum-only constraints to extract specific entities (e.g., identifying SFAS 131 major-customer links from 10-K Item 1).
3. Anonymizing text to neutralize the Glasserman-Lin distraction effect[51].
4. Requiring the LLM to output a verbatim citation string to ground its classification, mitigating hallucination.

4.4 Regime Switching vs. Optimal Execution and Sizing

REPORTED: The directive to mix strategies and switch on market phase is partially refuted by the literature. Hidden Markov Model (HMM) switchers frequently underperform buy-and-hold net of transaction costs[1]. The superior approach is **volatility targeting** and continuous risk scaling, which effectively truncates left-tail events without incurring the massive turnover drag of discrete regime switching.

Furthermore, execution costs scale violently with assets under management (AUM). The Almgren-Chriss (2000) optimal execution framework demonstrates that minimizing expected execution costs requires balancing permanent and temporary market impact against volatility risk[56]. At $1M AUM, a flat 6 bps cost model drastically understates the slippage of trading micro-caps. The square-root law of market impact ($I = Y \cdot \sigma \cdot \sqrt{Q/ADV}$) dictates that trading 2% clips of small-cap stocks can easily incur 60–75 bps in round-trip costs[1].

SPECULATION: Maximizing Return on Investment (ROI) over decades relies on maximizing the geometric growth rate $g = \mu - \sigma^2/2$. As volatility (σ) increases, the variance drag exponentially erodes the arithmetic return (μ). For a 19-year-old investor with decades of human capital, fractional Kelly sizing (e.g., half-Kelly) maximizes geometric growth while severely capping idiosyncratic tail risk. Concentrating capital into 10–12 micro-cap stocks is an uncompensated idiosyncratic risk; aggressive exposure should only be taken systematically across a diversified 40–50 stock book.

5. Strategic Roadmap: The 90-Day Execution Plan

The following ordered roadmap dictates exactly what to build, test, and kill, operating within a constrained budget and prioritizing maximum epistemic yield.

Phase 1: Data Integrity and Protocol Hardening (Days 1–15)

1. **Rectify the WRDS Supply Chain:** Re-pull the 109 MB BoardEx file removing the SQL LIMIT truncation. Transition all fundamental backtests from yfinance to the CRSP/Compustat Merged database. Utilize the crsp.ccmxpf_lnkhist table and strictly bound queries by linkdt and linkenddt to permanently resolve survivorship and CUSIP-reuse biases[24].
2. **Correct the Multiple-Testing Artifacts:** Recalculate the DSR of the 4 surviving strategies against the true denominator of $N = 179$ (incorporating parameter grids to reflect the true N). Discard strategies that fail to clear $t \geq 3.0$ or a RAS-adjusted Sharpe.
3. **Fix the Placebo Gate:** Rewrite the 13D/13G placebo algorithm to utilize a circular block bootstrap or strict permutation, eliminating the $\sqrt{5}$ pseudoreplication error[1]. **Kill Criterion:** If the seed-level t-statistic fails to remain significant under permutation

testing, the 13D/13G family must remain permanently closed.

Phase 2: Modernizing the Testing Infrastructure (Days 16–45)

4. **Integrate vectorbt and skfolio:** Replace the manual backtesting engine with vectorbt for highly accelerated signal generation[37]. Use skfolio to manage the portfolio assembly process, specifically leveraging its built-in cross-validation and Hierarchical Risk Parity (HRP) functionalities[34].
5. **Implement Hansen's SPA and Romano-Wolf:** Embed RSv618/superior-predictive-ability and the DoubleML implementation of the Romano-Wolf stepdown to create an un-gameable out-of-sample validation gate[10].
6. **Orthogonalize against openassetpricing:** Run all internal factors against Chen & Zimmermann's 319-characteristic dataset to verify that newly generated signals are not simply disguised manifestations of known factors[29].

Phase 3: The Context Thesis and LLM Extraction (Days 46–75)

7. **Build the Political & Network Access Signals:** Ingest public Senate LDA lobbying filings, federal contract data, and the newly corrected BoardEx linkages. Construct signals based on political alignment and corporate board interlocks, mirroring the findings of Faccio and Cooper et al.[1].
8. **Deploy Anonymized LLM Extraction:** Utilize the $20 DeepSeek budget to parse 10-K/10-Q filings for supply-chain dependencies and major customer data. Enforce strict prompt constraints (temperature=0, enum outputs) and anonymize all entities prior to parsing to bypass the distraction effect identified by Glasserman and Lin[50].
9. **Test the Neglect $\times$ Quality Interaction:** Cross high gross profitability with declining analyst coverage (ChNAnalyst from WRDS IBES) and low cosine-similarity news novelty. Measure the residual alpha after netting out transaction costs and short-availability constraints via the Almgren-Chriss impact models[46].

Phase 4: Synthesis, Publication, and Capital Allocation (Days 76–90)

10. **The Capital Allocation Pivot:** Discard the 10-stock mirror lane model. Re-allocate real capital using posterior-mean sizing across 40–50 names, weighted by inverse volatility or skfolio HRP to maximize geometric growth while capping idiosyncratic tail risk[34].
11. **Publish the Negative Results:** The primary marketable asset of this project is the ledger of 179 failures. Draft a methodology paper titled *"The Survival Curve of 179 Pre-Registered Candidate Signals"* detailing the placebo controls, the RAS implementations, and the multiple-testing deflations. Submit to the SSRN Financial Economics Network to establish a timestamped, peer-verifiable record.

By executing this roadmap, the Aegis-Finance project transitions from a fragile, heuristically guided framework into an institutionally rigorous, epistemically sound quantitative research apparatus capable of differentiating authentic market anomalies from statistical illusions.

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